Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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User Story:

- An IT organization receives employee data from external systems in the form of CSV or Excel files. The system administrator must import this data into ServiceNow and ensure that the relationships between employees, departments, and managers are correctly mapped.
- The application should automatically display the department, email, and manager information when an incident is assigned to an employee. This automation reduces manual work and improves data accuracy.

1. INTRODU CTION

√ Project Overview

- Thisprojectfocuses on automating the process of importing employee data intotheServiceNowplatform usingImportSets and Transform Maps.

- Organizations often maintain employee data in external spreadsheets or databases. Manually entering these records into the system is time-consuming and prone to errors.
- By using Import Sets in ServiceNow, employee records such as name, email, department, manager, and phone number can be imported automatically from external CSV files.
- Transform Maps help convert the imported data into the correct format and store it in the user table. Dot Walking allows related information such as department, manager, and email to be automatically displayed in incident records without manually opening each record.

√ **Purpose:**

❖ The main purpose of this project is to automate employee data import and improve relationship mapping between different records in ServiceNow.

❖ This system helps:

- Import employee data automatically
- Map relationships between users, departments, and managers
- Automatically display related information in incident forms
- Reduce manual effort and data entry errors

2. IDEATION PHASE

√ Problem Statement (Customer Problem Statement Template)

PS	I am	I'm trying to	but	Because	Which makes me feel
PS-1	System Administrator	Import employee data	It takes too much time	Data must be entered manually	Frustrated
PS-2	Incident Manager	View employee department & manager details	Information is not visible	Data is stored	Confused

√ Empathy Map Canvas

User Persona: System Administrator:

1. Think & Feel:

- Wants a faster way to import employee data and avoid manual errors.

2. Hear:

- Management expects quick system updates and accurate records.

3. See:

- Multiple spreadsheets containing employee information.

4. Say & Do:

- Manually imports files and checks data repeatedly.

5. <u>Pain:</u>

- Manual data entry takes time and increases the chance of mistakes.
-

6. <u>Gain</u>

- Automated data import and automatic relationship mapping.

√ **Brainstorming & Idea Prioritization:**

❖ Brainstorming is the process of generating ideas to solve a particular problem. During brainstorming, different approaches to automate data import were considered.

Some ideas discussed include:

- Manual data entry
- Automated import using scripts
- Using Import Sets in ServiceNow
- Mapping relationships using Dot Walking

√ After evaluating these options, Import Sets and Transform Maps were chosen as the most efficient solution.

Step 1 – Brainstormed Ideas

- Manual employee data entry
- Excel data management
- Automated data import using ServiceNow
- Integration with external systems

Step 2 – Grouping

- Manual methods
- Semi-automated systems
- Fully automated ServiceNow solution

Step 3 – Prioritization

- The ServiceNow-based automated solution using Import Sets and Dot Walking was selected because it provides efficient data management and reduces manual effort.

3. REQUIREMENT ANALYSIS PHASE

√ **Customer Journey Map**

- Stages Covered
- Employee data preparation in Excel
- Upload data into ServiceNow Import Sets
- Transform and map fields using Transform Map
- Store records in User table
- Create Incident and assign employee

- Automatically display department, email, and manager information

√ **Solution Requirements**

- Functional Requirements

FR No	Description
FR-1	Import employee data from CSV file
FR-2	Store employee records in ServiceNow User table
FR-3	Map employee fields using Transform Map
FR-4	Display department and manager using Dot Walking
FR-5	Show related information automatically in incident forms

Non-Functional Requirements

Usability

- The system should provide an easy method to import employee records without complex configurations.

Security

- Only authorized users such as administrators should be allowed to import or modify employee data.

Reliability

- The system must correctly transform and store all records without data loss.

Performance

- The system should process data quickly even when importing multiple records.

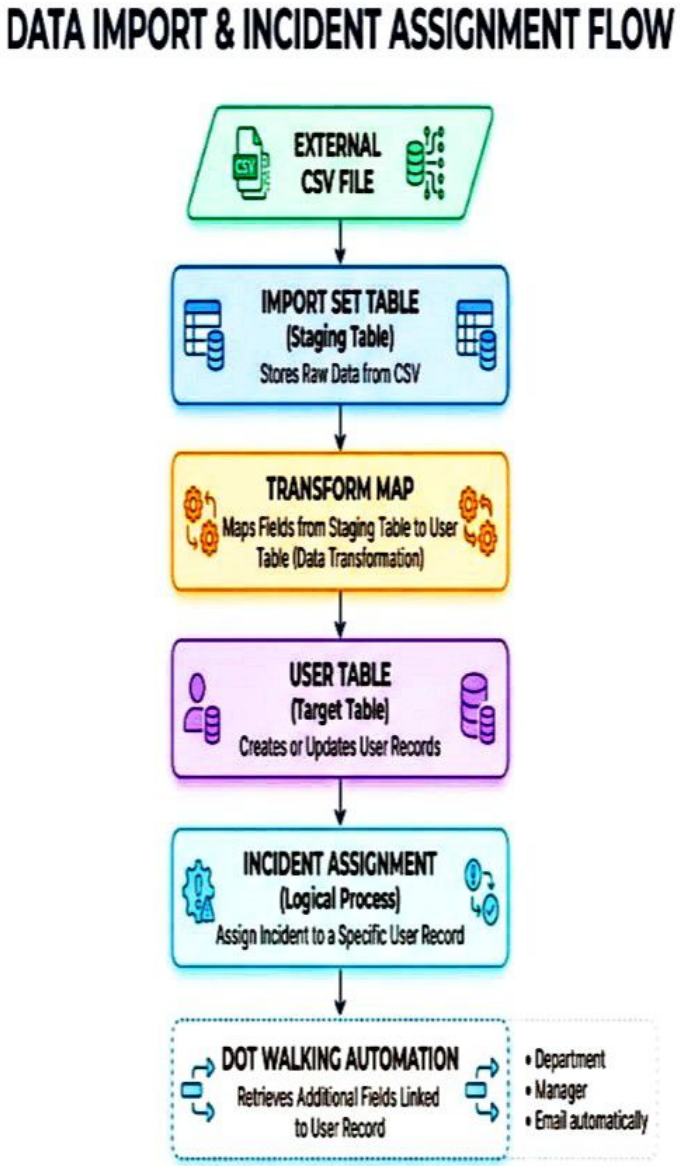
Scalability

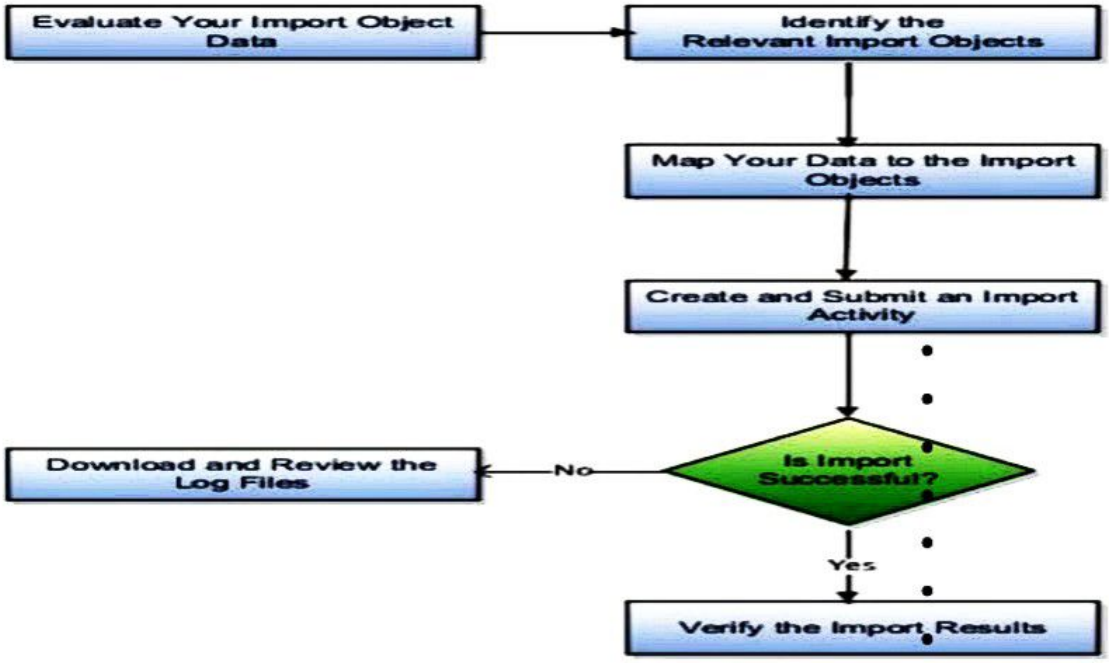
- The system should support large employee datasets in the future.

Availability

- The ServiceNow system should be accessible whenever administrators need to import or manage employee data.

√ **Data Flow Diagram:**





Entity Relationship Diagram (ER Diagram)

An Entity Relationship Diagram (ER Diagram) shows the relationship between different tables in the database.

In this project, the main entities are:

User

Department

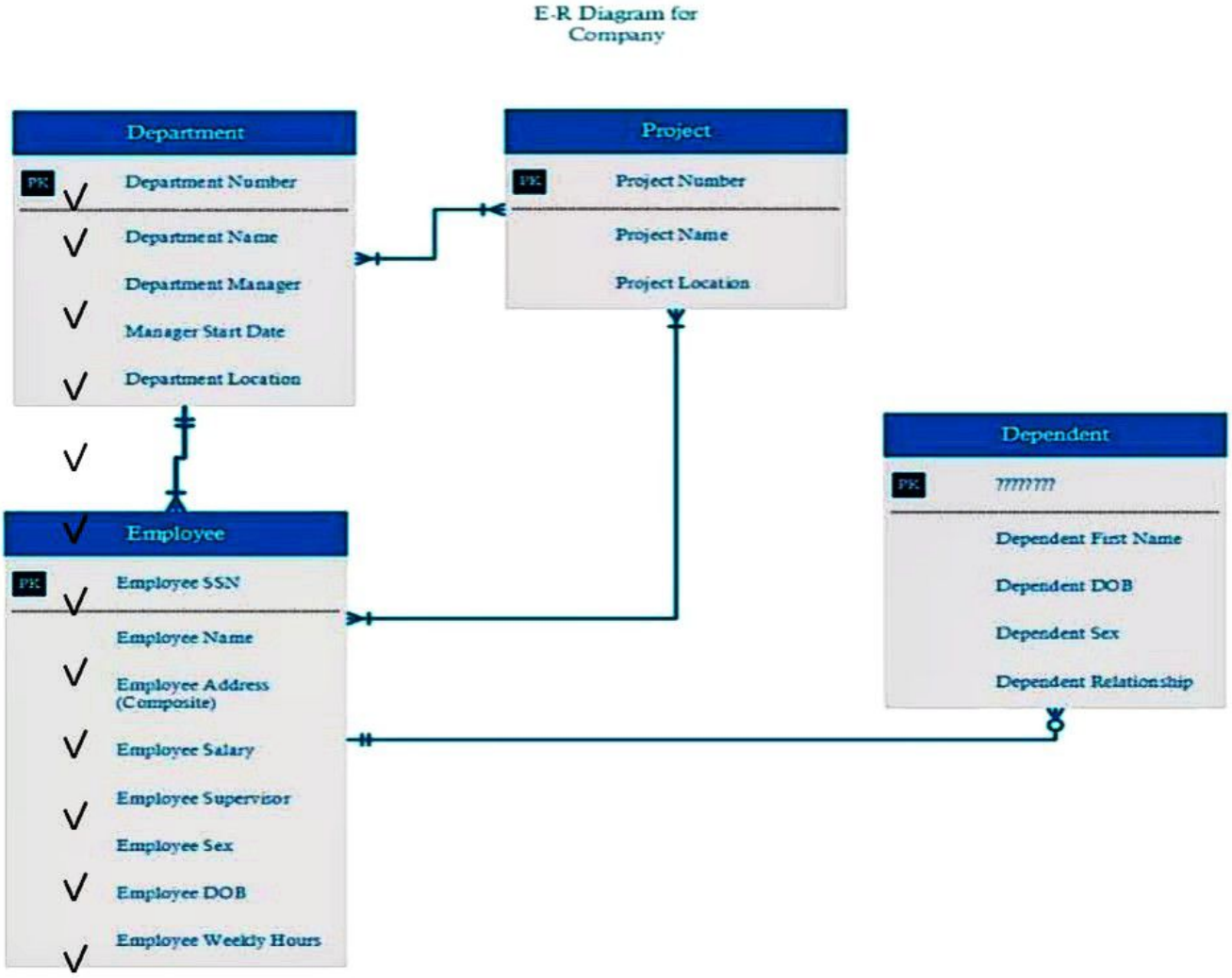
Manager

Incident

The User table stores employee details such as name, email, department, and manager. The Department table stores information about different departments in the organization.

Each user belongs to a department and may have a manager. These relationships are maintained within the User table.
The Incident table stores incidents reported in the system. Each incident is assigned to a user.

The ER diagram helps in understanding the structure of the database and how different tables are connected.



• Security and Access Control

Security is an important aspect of any information system. In ServiceNow, security is implemented using Role-Based Access Control (RBAC).

Different users are given different levels of permissions based on their roles.

For example:

Admin – Full system access

IT Staff – Access to incident management

Employees – Limited access to report incidents

This ensures that only authorized users can access or modify system data.

√ Error Handling

During the data import process, some errors may occur due to incorrect data or missing values.

Examples of common errors include:

Incorrect data format

Missing mandatory fields

Duplicate records

ServiceNow Import Sets provide error logs during the transformation process. These logs help administrators identify and correct errors.

Proper error handling ensures that the system maintains accurate and reliable data.

√ Technology Stack

Layer	Technology
Platform	ServiceNow
Data Source	CSV / Excel file
Import Tool	Import Sets
Data Mapping	Transform Maps
Relationship Access	Dot Walking

4. PROJECT DESIGN PHASE

√ Problem – Solution Fit

The project solves the problem of manual employee data entry by automating the import process and mapping relationships between records using ServiceNow features.

√ Proposed Solution

Parameter	Description
Problem	Manualdata entry ofemployee records
Solution	Automated import using Import Sets
Novelty	Automatic relationship mapping
Social Impact	Faster IT service management
Scalability	Supports large employee data

5. SYSTEM ARCHITECTURE

System architecture describes how different components of the system interact with each other.

The architecture of this project consists of the following components:

External Data Source

Import Set Table

Transform Map

Target Table

Dot Walking for relationship access

The external data source contains the raw data which needs to be imported. This data is uploaded into ServiceNow using the Load Data module. ServiceNow creates an Import Set table where the data is temporarily stored.

Transform Maps define how the data should be transferred from the import set table to the target table. Once the transform process is executed, the data is inserted into the target table.

Dot Walking allows users to access related data from referenced tables. This helps users retrieve important information quickly.

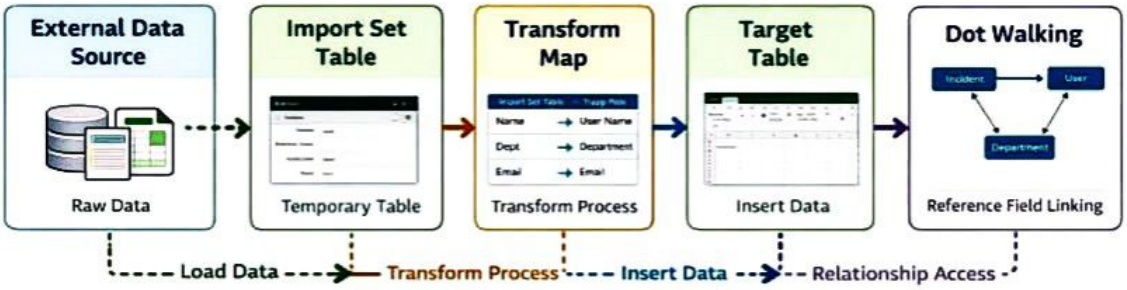


Figure: System Architecture of Data Import Process

6. PROJECT DEVELOPMENT PHASE

Project Flow

Milestone 1 – Create Employee Data in Excel

Milestone 2 – Import Data using Import Sets

Milestone 3 - Create Transform Map

Milestone 4 - Map Fields using Mapping Assist

Milestone 5 - Transform Data into User Table

Milestone 6 - Use Dot Walking in Incident Form

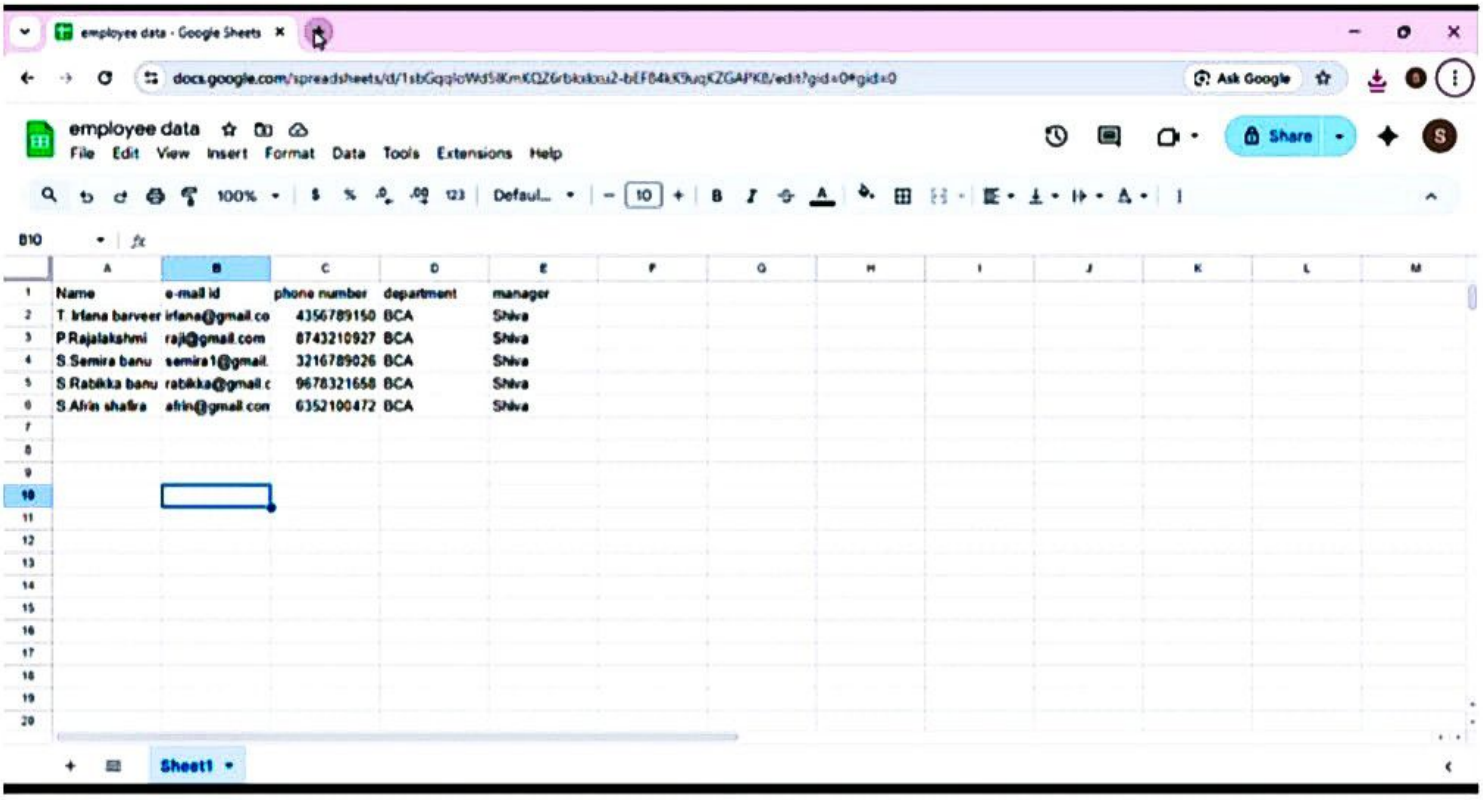
Milestone 7 - Display Department, Email, and Manager Automatically

7.IMPLEMENTATION

Implementation describes how the system was developed using ServiceNow tools.

Step 1: Uploading Data

Data is uploaded into ServiceNow using the Load Data module.



Name	e-mail id	phone number	department	manager
T Irfana barveer	irfana@gmail.co	4356789150	BCA	Shiva
P Rajalakshmi	raji@gmail.com	8743210927	BCA	Shiva
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S Rabikka banu	rabikka@gmail.c	9678321658	BCA	Shiva
S Alvin shafira	alvin@gmail.com	6352100472	BCA	Shiva

Figure: Uploading CSV File into ServiceNow

Step 2: Import Set Table Creation

After uploading the file, ServiceNow automatically creates an Import Set table.

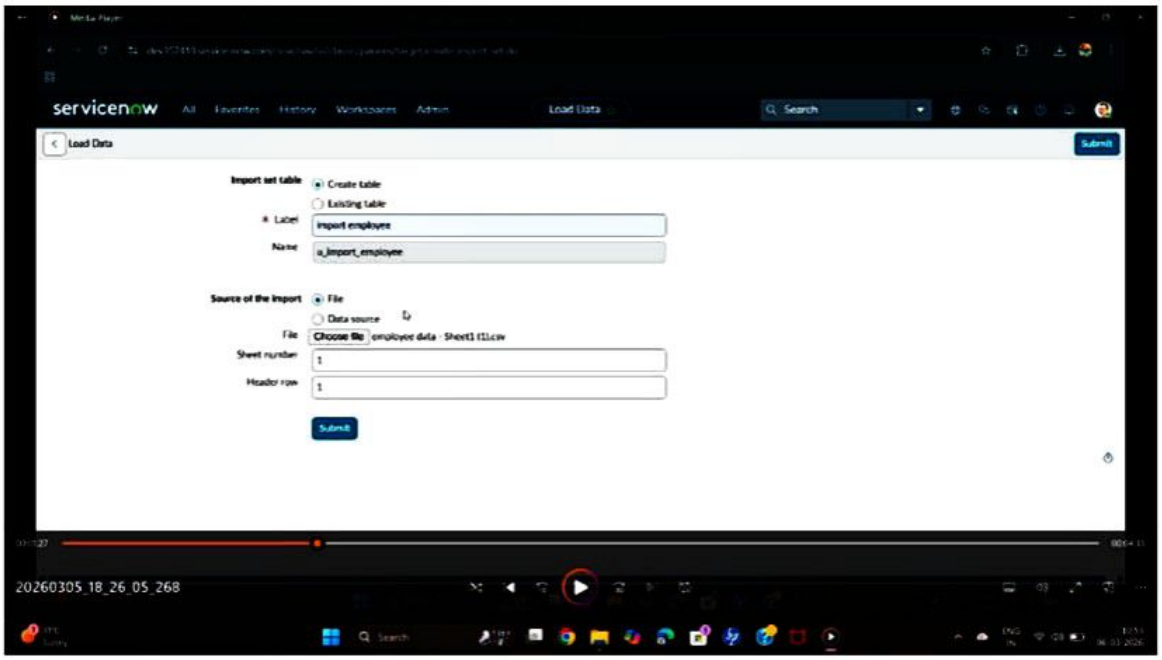

Figure: Import Set Table with Imported Data

Step 3: Creating Transform Map

Transform Maps are used to map fields between the import set table and the target table.

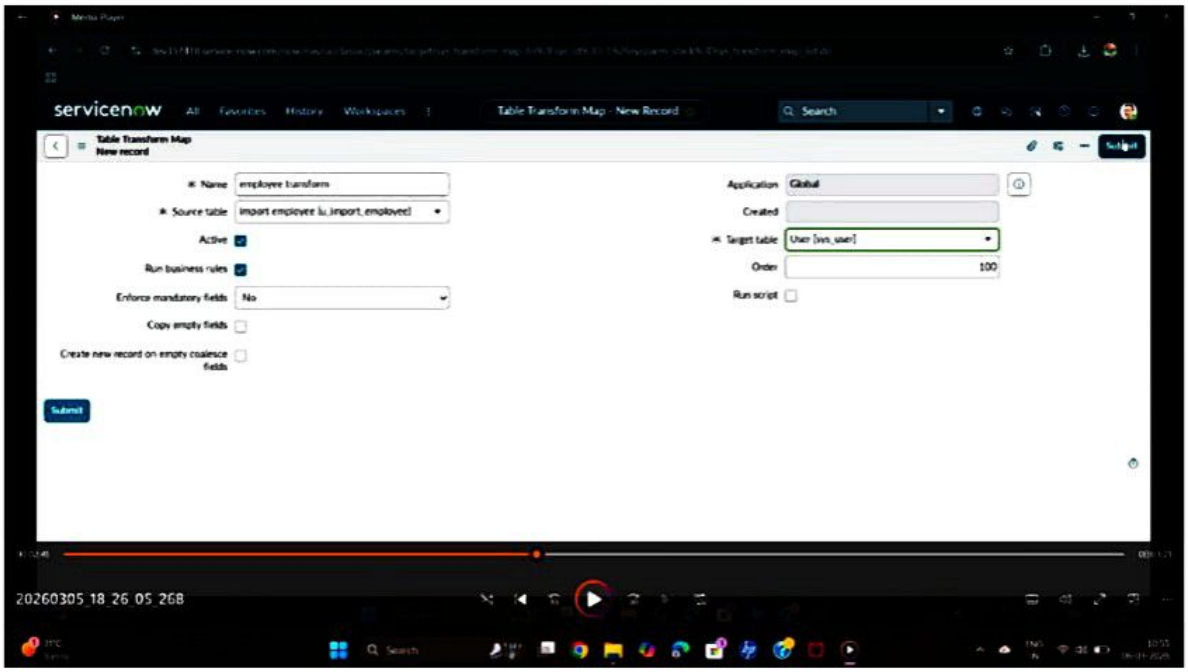

Figure: Transform Map Configuration

Step 4: Field Mapping

Fields from the source table are mapped to fields in the target table.

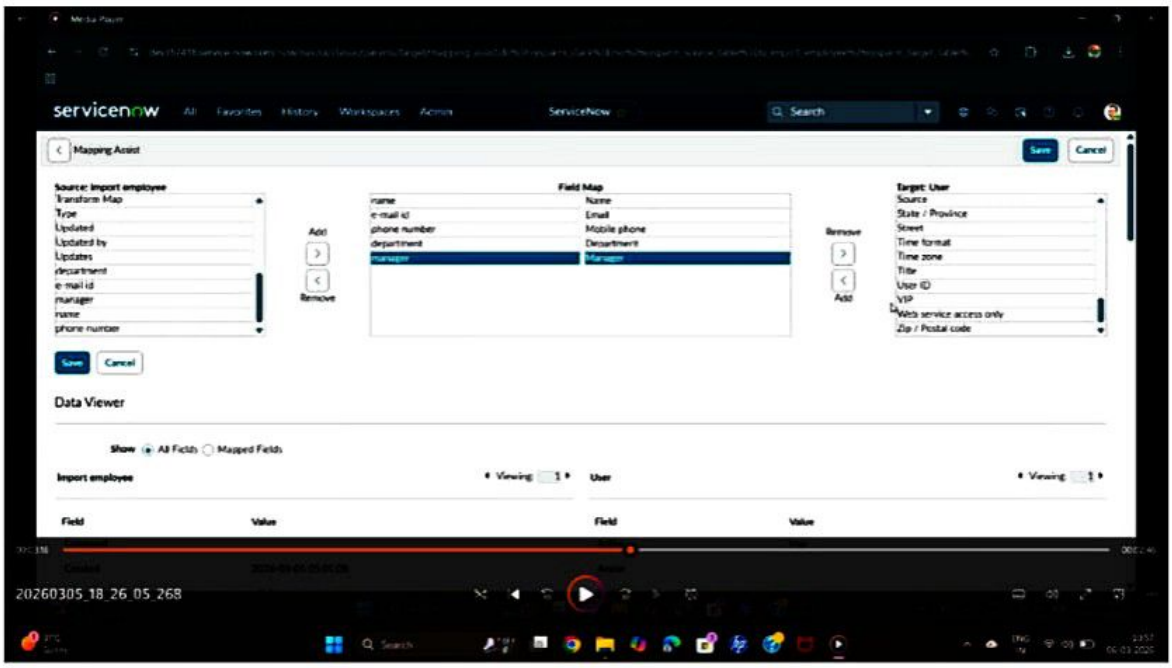

Figure: Source and Target Field Mapping

Step 5: Running Transform Map

The transform map is executed to transfer data from the import set table to the target table.

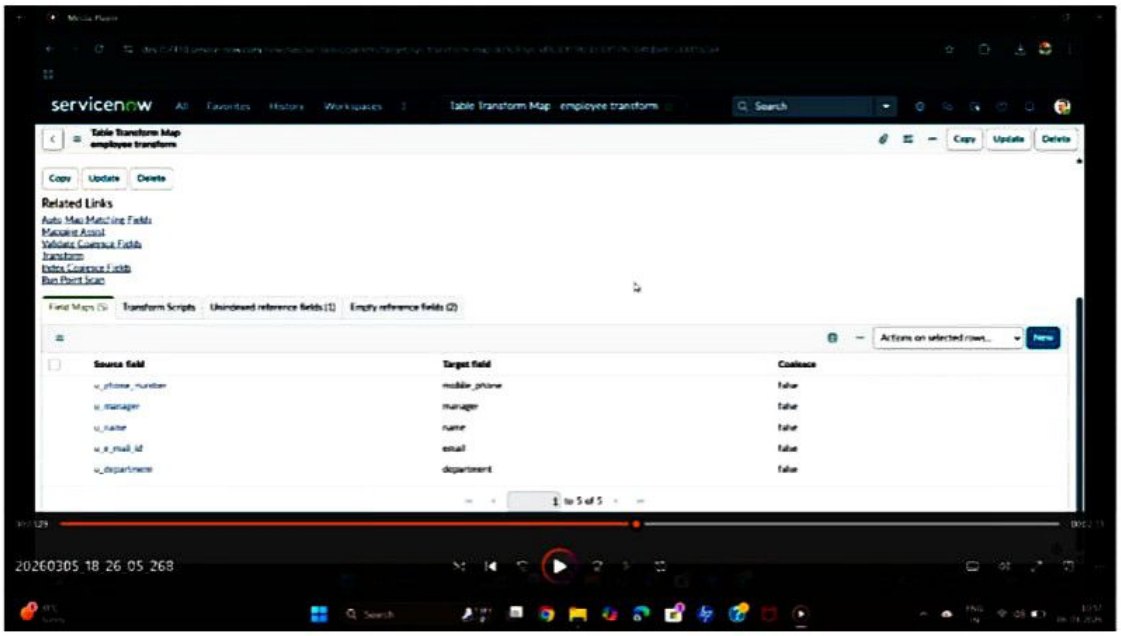

Figure: Transform Execution Result

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Step 6: Viewing Final Data

After transformation, the data appears in the target table.

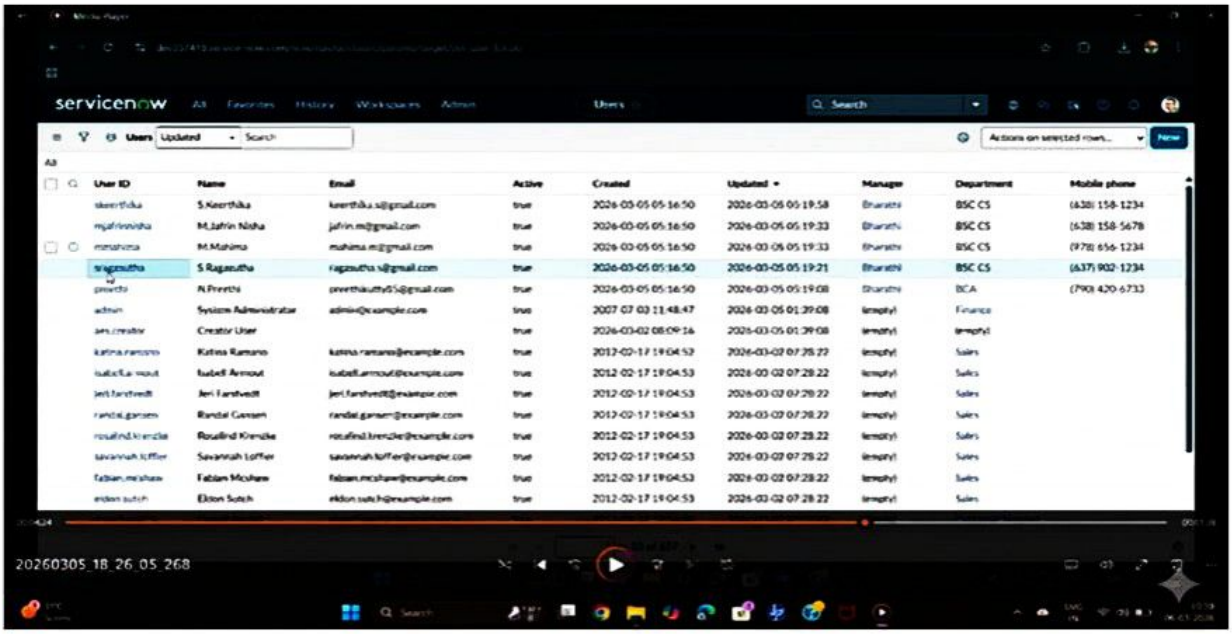


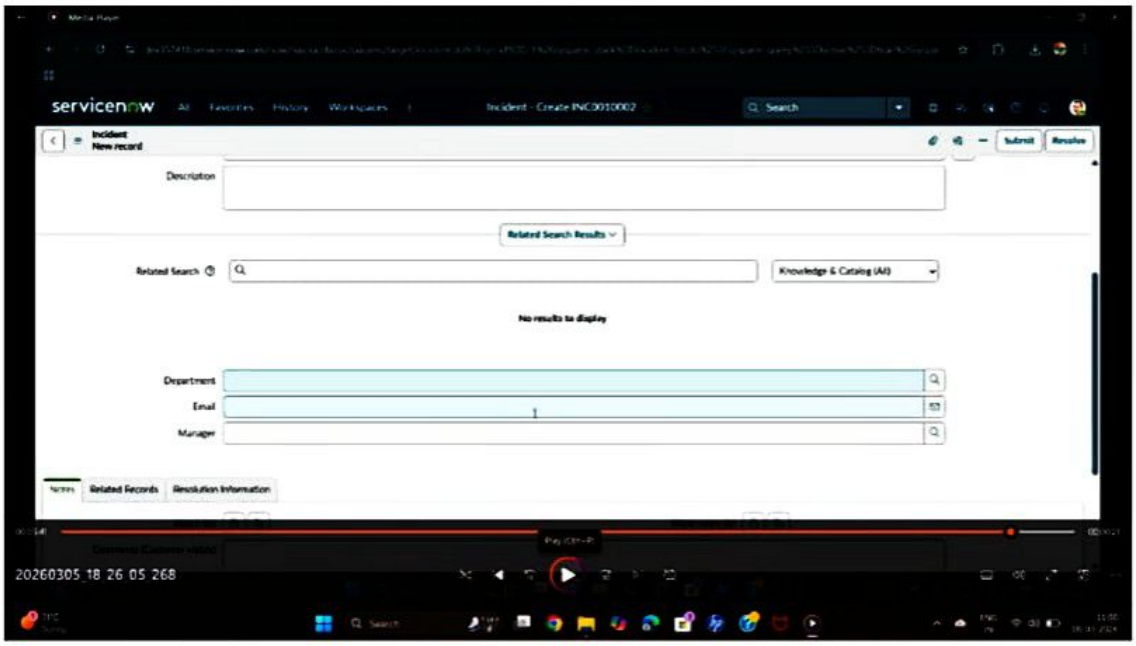
Figure: Imported Records in Target Table

8. RESULTS

The project successfully demonstrated how data can be imported automatically using Import Sets.

The results show that:

- Data import time is reduced
- Manual effort is minimized
- Data relationships are maintained
- Accuracy of data improves



9. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual data entry
- Improves data accuracy
- Saves time for administrators
- Automatically displays related information

Disadvantages

- Requires ServiceNow knowledge
- Initial setup is required for mapping field.

9. Limitations of the System

Although the system provides automation and efficiency, it has certain limitations.

Some limitations include:

Importing very large data files may take more time

Incorrect data formats may cause errors

Manual verification may sometimes be required

However, these limitations can be minimized through proper data preparation and system configuration.

10. CONCLUSION

This project demonstrates how ServiceNow can automate employee data import and relationship mapping using Import Sets, Transform Maps, and Dot Walking.

The system improves efficiency by automatically importing employee data and displaying related information such as department, email, and manager details when incidents are assigned.

11. FUTURE ENHANCEMENTS

- Integration with HR management systems
- Automatic employee record updates
- Advanced reporting and dashboards
- Real-time system integration